Data Science Demo

A short two-page overview showing how data science turns raw data into decisions.

What is data science?

Data science combines statistics, programming, domain knowledge, and communication to answer practical questions using data. The goal is not only to build models, but also to understand patterns, reduce uncertainty, and support better decisions.

Typical workflow

Step	Purpose	Example Output
1. Ask	Define the business or research question.	Predict churn, estimate demand, detect fraud
2. Collect	Gather structured and unstructured data.	CSV files, logs, surveys, APIs
3. Clean	Fix missing values, duplicates, and inconsistent types.	Clean analysis table
4. Explore	Use summaries and charts to understand the data.	Trends, outliers, correlations
5. Model	Train and evaluate a statistical or ML model.	Regression, classifier, forecast
6. Communicate	Explain results clearly for decisions.	Report, dashboard, recommendation

Key roles in a data science project

Role	Main contribution
Data Engineer	Builds reliable data pipelines and storage.
Data Analyst	Explores data and explains trends.
Machine Learning Engineer	Trains, deploys, and monitors models.
Domain Expert	Checks whether conclusions make real-world sense.

Mini example

Question: Which marketing channel produced the strongest monthly sales? The data scientist would compare channels, check anomalies, and recommend where to focus budget.

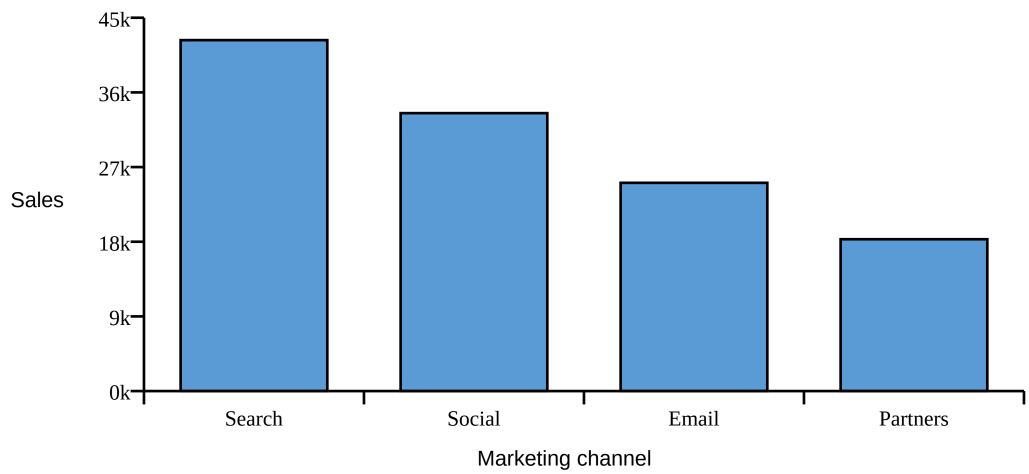
Simple Data Science Mini-Report

This page uses a small fictional dataset to illustrate exploratory analysis and model evaluation.

Example dataset

Channel	January	February	March	Total
Search Ads	12000	14500	15800	42300
Social Media	9000	11000	13500	33500
Email	7000	8500	9600	25100
Partners	5000	6100	7200	18300

Total Sales by Channel



Interpretation

Search Ads has the highest total sales in this sample. Social Media is second and shows a strong month-to-month increase. A practical next step would be to compare sales against cost, because the best channel is the one with the strongest return on investment, not only the highest revenue.

Simple model evaluation

Metric	Meaning	Demo Value
Accuracy	Share of predictions that were correct.	0.86
Precision	How many predicted positives were truly positive.	0.82
Recall	How many true positives were found by the model.	0.79
F1-score	Balance between precision and recall.	0.80